

HNSS18 · Eye & Ear Developmental Abnormalities — Pathology Chart

Each defect traced to the developmental step that failed | **STEP 1** = beyond the lecture · **LOWER PRIORITY** = not in this year's lecture · **EXAM**
EMPHASIS = flagged in class as testable

HoyaDocs
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The Eye						
DEFECT	WHO / CAUSE	WHAT FAILED (EMBRYOLOGY)	KEY FINDINGS	GENE / MOLECULAR	DIAGNOSIS & BUZZWORDS	MANAGEMENT & KEY FACTS
Cyclopia <i>the eye fields that never split</i> EYE FIELDS	Loss of SHH signaling; holoprosencephaly risks: SHH mutation, trisomy 13 , maternal diabetes, alcohol STEP 1	SHH from the anterior ventral midline normally represses PAX6 in the midline and splits the eye field in two; without it the fields converge	Single midline eye; forebrain deficient	SHH (loss) · PAX6 (eye-field gene)	On the holoprosencephaly spectrum STEP 1	Severe; usually incompatible with life STEP 1
Coloboma (iris / uveal) <i>the fissure that didn't close</i> CHOROID FISSURE	Congenital	Choroid fissure fails to close (the inferior groove in the optic cup and stalk that carries the hyaloid artery, ~wk 6–7)	Inferior “ keyhole ” gap in the iris; can extend to the retina, choroid, optic nerve	PAX2 (renal-coloboma) STEP 1	Keyhole pupil ; part of CHARGE (Coloboma, Heart, Atresia choanae, Retardation, GU, Ear) STEP 1	Don't confuse with the lower-eyelid coloboma of Treacher Collins (arch 1 neural crest)
Persistent pupillary membrane <i>the curtain that didn't open</i> PUPILLARY MEMBRANE	Congenital	The pupillary membrane (iridopupillary membrane in front of the lens) fails to regress before birth	Strands of tissue across the pupil	—	Usually harmless; matters when extensive	If it obstructs the visual axis → deprivation amblyopia
Aniridia <i>the missing iris</i> PAX6 STEP 1	PAX6 mutation (sporadic or AD)	Abnormal development of the anterior rim of the optic cup (iris)	Absent / hypoplastic iris ; photophobia, ↓ vision	PAX6 (11p13)	WAGR : Wilms tumor, Aniridia, GU anomalies, intellectual disability	Screen for Wilms tumor in WAGR
Congenital cataract <i>the cloudy lens</i> LENS STEP 1	In-utero rubella ; also galactosemia	Lens fiber injury during development (surface ectoderm–derived lens)	Cloudy lens, absent red reflex / leukocoria	—	Congenital rubella triad: cataracts, sensorineural deafness, PDA	Early surgery to prevent amblyopia

★ **USMLE high-yield: Coloboma = choroid fissure** failed to close (inferior keyhole iris). **Cyclopia = no SHH** → PAX6 not repressed → fields fuse. **PAX6 = master eye gene** (aniridia, WAGR). Leukocoria in a newborn → think congenital cataract (rubella); in a child → **retinoblastoma** (RB1).

The Ear						
DEFECT	WHO / CAUSE	WHAT FAILED (EMBRYOLOGY)	KEY FINDINGS	GENE / MOLECULAR	DIAGNOSIS & BUZZWORDS	MANAGEMENT & KEY FACTS
Low-set / malformed ears <i>the ears that didn't rise</i> AURICLE	Arch 1–2 syndromes; chromosomal disorders (trisomy 18, 21) STEP 1	Abnormal auricular hillocks (1–3 arch 1, 4–6 arch 2), or poor jaw growth fails to move the ears up from their caudal start	Low-set, small or misshapen auricles (microtia)	—	Ear position relative to eye and nose on fetal ultrasound (HNSS04)	A marker of an underlying syndrome — external ear deformity → search for other pharyngeal-arch malformations EXAM EMPHASIS
Drug-induced ear anomalies <i>the first-trimester ear</i> EXTERNAL EAR LOWER PRIORITY	First-trimester streptomycin , salicylates , thalidomide	External ear is very sensitive to drug exposure during formation	Pharyngeal and external ear malformations	—	Aminoglycosides (streptomycin) are also ototoxic → sensorineural loss STEP 1	Avoid in pregnancy
Waardenburg syndrome <i>the pigment-and-hearing crest defect</i> NEURAL CREST	Autosomal dominant	Neural crest defect affecting pigment cells (melanocytes, incl. those in the cochlea)	Congenital deafness + white forelock , heterochromia, wide-set eyes	PAX3	White forelock + deafness	Sensorineural hearing loss
Genetic hair-cell loss <i>the broken receptors</i> INNER EAR LOWER PRIORITY	Inherited	Defects in the otic-vesicle–derived organ of Corti and its support	Sensorineural hearing loss	POU4F3 (hair-cell degeneration) · connexin 26 (supporting-cell gap junctions) · c-Kit (stria vascularis)	Connexin 26 (GJB2) = most common genetic cause of congenital deafness STEP 1	Hearing aids / cochlear implants STEP 1
Treacher Collins syndrome <i>the first arch that didn't fill in</i> ARCH 1 NC STEP 1	Autosomal dominant; Tcof-1 (HNSS03)	NC from R2/R4 fails to migrate into arches 1–2 → malleus/incus and auricle malformed	Malformed external ears, small jaw and cheekbones, lower-eyelid colobomas	Tcof-1	Conductive hearing loss (ossicles / canal)	Hearing aids, reconstruction
Congenital infection deafness <i>the infected inner ear</i> INNER EAR STEP 1	In-utero CMV or rubella	Infection damages the developing inner ear	Sensorineural hearing loss (± cataracts, PDA with rubella)	—	CMV = most common infectious cause of congenital hearing loss	Newborn hearing screen

Ear histology lecture — exam pearls (not embryology, listed so they aren't missed)

Otosclerosis EXAM EMPHASIS	Overgrowth/fixation of the ossicles (usually the stapes) → conductive loss; worsened by pregnancy hormones → pregnant woman with new hearing loss = otosclerosis
Cochlear implant EXAM EMPHASIS	Electrodes stimulate the cochlea by frequency position — works only if the auditory nerve (CN VIII) is functional
Genetic hair-cell loss	Connexin 26 (gap junctions, most common genetic deafness) · Pou4f3 · c-Kit (stria vascularis)

★ **USMLE high-yield:** Inner ear = **otic placode** (surface ectoderm) → defects cause **sensorineural** loss; middle/external ear = **pouch 1, cleft 1, arches 1–2** → defects cause **conductive** loss. Low-set ears = arch 1–2 or chromosomal marker.